



**18th Iranian Statistics Conference**  
**K.N.Toosie University of Thechnology**

**29-31 August 2026**



## **An unified view of probabilistic inference under incomplete information: Maximum Entropy and Bayesian Updating .**

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### **Abstract:**

Statistical inference under incomplete information involves two complementary but distinct tasks: assigning an initial probability distribution when knowledge is limited to partial constraints, and updating that distribution when observations become available. Maximum Entropy addresses the first task by selecting a distribution consistent with the known moment conditions, thereby avoiding assumptions unsupported by the available information. Bayesian inference addresses the second by using the likelihood to revise prior probabilities in light of observed data. Recent work connects these roles within a single Bayesian framework: moment constraints determine the prior weighting of candidate distributions, while observations modify those weights through the likelihood. When the constraint-based prior is sharply concentrated around the entropy-maximizing distribution, the Bayesian result approaches the Maximum Entropy solution. This review examines this connection and illustrates the resulting probabilistic updating process through a numerical experiment.

**Keywords:** Maximum Entropy, Bayesian Inference, and Entropic Priors.

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